

Neural Network and Deep Learning

Big Data Analytics

Readings

- Neural Network and Deep Learning
 - Chap. 1

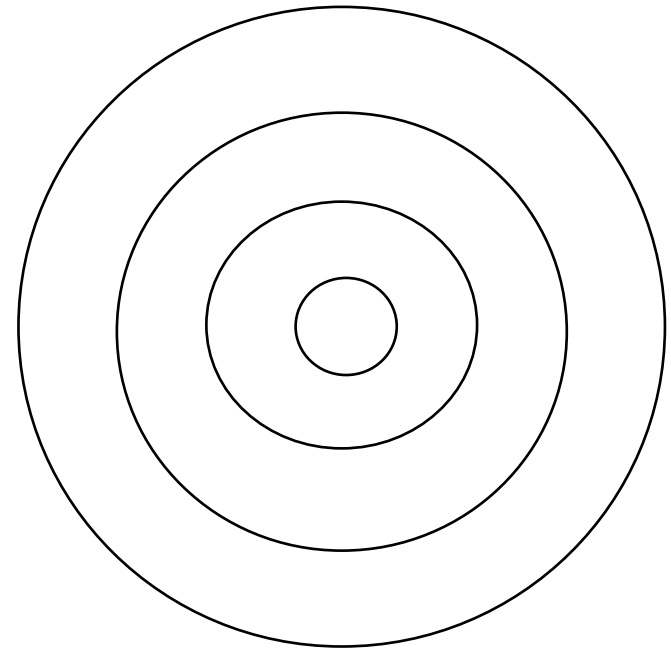
Mini Batch Gradient Descent

- Estimate the gradient ∇C by computing ∇C_{x_j} for a small sample of randomly chosen training inputs $X_1, X_2, \dots, X_m$:
 - Size m is large enough to expect that the average value of ∇C_{X_j} will be roughly equal to the average over all ∇C_x .

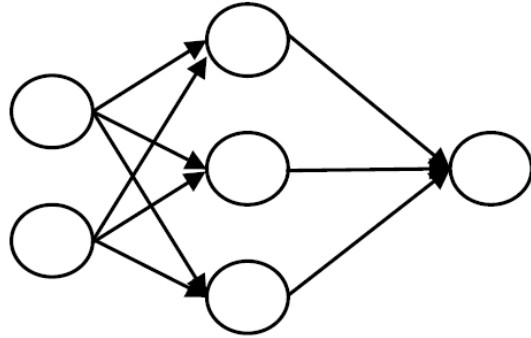
$$\frac{\sum_{j=1}^m \nabla C_{X_j}}{m} \approx \frac{\sum_x \nabla C_x}{n} = \nabla C \quad \nabla C \approx \frac{1}{m} \sum_{j=1}^m \nabla C_{X_j}$$

$$w_k \rightarrow w'_k = w_k - \frac{\eta}{m} \sum_j \frac{\partial C_{X_j}}{\partial w_k}$$

$$b_l \rightarrow b'_l = b_l - \frac{\eta}{m} \sum_j \frac{\partial C_{X_j}}{\partial b_l},$$



Below is a neural network:



All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{x}_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$\mathbf{y}_1 = \begin{bmatrix} 0 \end{bmatrix} \quad \mathbf{y}_2 = \begin{bmatrix} 0 \end{bmatrix} \quad \mathbf{y}_3 = \begin{bmatrix} 1 \end{bmatrix} \quad \mathbf{y}_4 = \begin{bmatrix} 1 \end{bmatrix}$$

And the learning rate is $\eta = 0.6$

We are processing 4 inputs at the same time by having all 4 of them as a matrix.

Round off all the number to the nearest 1000th. For example, 6.6543 is 6.654 and 3.3355 is 3.336.

z	-0.262	-0.362	-0.462	-0.962	-1.133	0.745	1.233	1.425	1.518
a	0.435	0.410	0.387	0.276	0.244	0.678	0.774	0.806	0.820

z	0	0.5	1	1.5	2	-0.5	-1	-1.5	-2
a	0.5	0.622	0.731	0.818	0.881	0.378	0.269	0.182	0.119

And the back propagation equations are as follows:

Summary: the equations of backpropagation

$$\delta^L = \nabla_a C \odot \sigma'(z^L) \quad (\text{BP1})$$

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l) \quad (\text{BP2})$$

$$\frac{\partial C}{\partial b_j^l} = \delta_j^l \quad (\text{BP3})$$

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l \quad (\text{BP4})$$

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 1 \end{bmatrix}$$

(a) [4 Points] What is Z^2 matrix? First represent it with matrix format and write the result. Show all your work.

And the back propagation equations are as follows:

Summary: the equations of backpropagation

$$\delta^L = \nabla_a C \odot \sigma'(z^L) \quad (\text{BP1})$$

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l) \quad (\text{BP2})$$

$$\frac{\partial C}{\partial b_j^l} = \delta_j^l \quad (\text{BP3})$$

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l \quad (\text{BP4})$$

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [0] \quad y_3 = [1] \quad y_4 = [1]$$

(a) [4 Points] What is Z^2 matrix? First represent it with matrix format and write the result. Show all your work.

$$\begin{aligned} Z^2 &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} + \begin{bmatrix} -1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & -1 \end{bmatrix} + \begin{bmatrix} -1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix} \end{aligned}$$

(b) [4 Points] What is a^2 matrix? Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 1 \end{bmatrix}$$

(c) [4 Points] What is Z^3 matrix? First represent it with matrix format and write the result. Show all your work.

(b) [4 Points] What is a^2 matrix? Show all your work.

$$a^2 = \begin{bmatrix} 0.269 & 0.269 & 0.5 & 0.5 \\ 0.5 & 0.731 & 0.5 & 0.731 \\ 0.731 & 0.731 & 0.5 & 0.5 \end{bmatrix}$$

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 1 \end{bmatrix}$$

(c) [4 Points] What is Z^3 matrix? First represent it with matrix format and write the result. Show all your work.

(b) [4 Points] What is a^2 matrix? Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 1 \end{bmatrix}$$

$$a^2 = \begin{bmatrix} 0.269 & 0.269 & 0.5 & 0.5 \\ 0.5 & 0.731 & 0.5 & 0.731 \\ 0.731 & 0.731 & 0.5 & 0.5 \end{bmatrix}$$

(c) [4 Points] What is Z^3 matrix? First represent it with matrix format and write the result. Show all your work.

$$Z^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} a^2 + \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\approx \begin{bmatrix} -0.462 & -0.462 & 0 & 0 \end{bmatrix}$$

(d) [4 Points] What is a^3 matrix? Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [0] \quad y_3 = [1] \quad y_4 = [1]$$

(e) [3 Points] What is ${}^1\delta^3$ value? ${}^1\delta^3$ is the layer 3 delta for input 1. Show all your work.

(d) [4 Points] What is a^3 matrix? Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [0] \quad y_3 = [1] \quad y_4 = [1]$$

$$a^3 = [0.387 \quad 0.387 \quad 0.5 \quad 0.5]$$

(e) [3 Points] What is ${}^1\delta^3$ value? ${}^1\delta^3$ is the layer 3 delta for input 1. Show all your work.

(d) [4 Points] What is a^3 matrix? Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 1 \end{bmatrix}$$

$$a^3 = \begin{bmatrix} 0.387 & 0.387 & 0.5 & 0.5 \end{bmatrix}$$

(e) [3 Points] What is ${}^1\delta^3$ value? ${}^1\delta^3$ is the layer 3 delta for input 1. Show all your work.

$${}^1\delta^3 = \begin{bmatrix} 0.387 \times 0.387 \times (1 - 0.387) \end{bmatrix} = \begin{bmatrix} 0.092 \end{bmatrix}$$

(f) [3 Points] What is ${}^2\delta^3$ value? ${}^2\delta^3$ is the layer 3 delta for input 2. Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [0] \quad y_3 = [1] \quad y_4 = [1]$$

(g) [3 Points] What is ${}^3\delta^3$ value? ${}^3\delta^3$ is the layer 3 delta for input 3. Show all your work.

(f) [3 Points] What is ${}^2\delta^3$ value? ${}^2\delta^3$ is the layer 3 delta for input 2. Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [0] \quad y_3 = [1] \quad y_4 = [1]$$

$${}^2\delta^3 = [0.092]$$

(g) [3 Points] What is ${}^3\delta^3$ value? ${}^3\delta^3$ is the layer 3 delta for input 3. Show all your work.

(f) [3 Points] What is ${}^2\delta^3$ value? ${}^2\delta^3$ is the layer 3 delta for input 2. Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [0] \quad y_3 = [1] \quad y_4 = [1]$$

$${}^2\delta^3 = [0.092]$$

(g) [3 Points] What is ${}^3\delta^3$ value? ${}^3\delta^3$ is the layer 3 delta for input 3. Show all your work.

$${}^3\delta^3 = [-0.5 \times 0.5 \times 0.5] = [-0.125]$$

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 1 \end{bmatrix}$$

(h) [3 Points] What is δ^3 value? δ^3 is the layer 3 delta for input 4. Show all your work.

(i) [4 Points] What is average $\frac{\partial C}{\partial b^3}$ vector? Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 1 \end{bmatrix}$$

(h) [3 Points] What is δ^3 value? δ^3 is the layer 3 delta for input 4. Show all your work.

$$\delta^3 = \{-0.125\}$$

(i) [4 Points] What is average $\frac{\partial C}{\partial b^3}$ vector? Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 1 \end{bmatrix}$$

(h) [3 Points] What is δ^3 value? δ^3 is the layer 3 delta for input 4. Show all your work.

$$\delta^3 = \{-0.125\}$$

(i) [4 Points] What is average $\frac{\partial C}{\partial b^3}$ vector? Show all your work.

$$\begin{aligned} \frac{\partial C}{\partial b^3} &= [(0.092 + 0.092 - 0.125 - 0.125) / 4] \\ &= [-0.017] \end{aligned}$$

(j) [12 Points] What is average $\frac{\partial C}{\partial w^3}$ matrix? Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 1 \end{bmatrix}$$

(j) [12 Points] What is average $\frac{\partial C}{\partial w^3}$ matrix? Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 1 \end{bmatrix}$$

$$\frac{\partial C}{\partial w_{11}^3} = \left[\frac{(0.269 \times 0.092 + 0.269 \times 0.092 - 0.5 \times 0.125 - 0.5 \times 0.125)}{4} \right] = [-0.019]$$

$$\frac{\partial C}{\partial w_{12}^3} = \left[\frac{(0.5 \times 0.092 + 0.731 \times 0.092 - 0.5 \times 0.125 - 0.731 \times 0.125)}{4} \right] = [-0.010]$$

$$\frac{\partial C}{\partial w_{13}^3} = \left[\frac{(0.731 \times 0.092 + 0.731 \times 0.092 - 0.5 \times 0.125 - 0.5 \times 0.125)}{4} \right] = [0.002]$$

$$\frac{\partial C}{\partial w^3} = \begin{bmatrix} -0.019 & -0.010 & 0.002 \end{bmatrix}$$

All the matrices and vectors are as follows:

$$W^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad W^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [0] \quad y_3 = [1] \quad y_4 = [1]$$

(k) [3 Points] What is new b^3 vector? Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [0] \quad y_3 = [1] \quad y_4 = [1]$$

(k) [3 Points] What is new b^3 vector? Show all your work.

$$b^3 \rightarrow b^{3'} = [0] - 0.6 [-0.017] = [0.0102]$$

(I) [3 Points] What is new w^3 matrix? Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [0] \quad y_3 = [1] \quad y_4 = [1]$$

(I) [3 Points] What is new w^3 matrix? Show all your work.

All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [0] \quad y_3 = [1] \quad y_4 = [1]$$

$$\begin{aligned} w^3 \rightarrow w^{3'} &= \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} - 0.6 \begin{bmatrix} -0.019 & -0.010 \\ & 0.002 \end{bmatrix} \\ &= \begin{bmatrix} 1.011 & 0.006 & -1.001 \end{bmatrix} \end{aligned}$$

- Proof of BP1 $\delta_j^L = \frac{\partial C}{\partial a_j^L} \sigma'(z_j^L)$

- Proof of BP1

$$\delta_j^L = \frac{\partial C}{\partial a_j^L} \sigma'(z_j^L)$$

$$\delta_j^L = \frac{\partial C}{\partial z_j^L}$$

- Proof of BP1 $\delta_j^L = \frac{\partial C}{\partial a_j^L} \sigma'(z_j^L)$

$$\delta_j^L = \frac{\partial C}{\partial z_j^L}$$

$$\delta_j^L = \sum_k \frac{\partial C}{\partial a_k^L} \frac{\partial a_k^L}{\partial z_j^L}$$

- Proof of BP1 $\delta_j^L = \frac{\partial C}{\partial a_j^L} \sigma'(z_j^L)$

$$\delta_j^L = \frac{\partial C}{\partial z_j^L}$$

$$\delta_j^L = \sum_k \frac{\partial C}{\partial a_k^L} \frac{\partial a_k^L}{\partial z_j^L}$$

$$\delta_j^L = \frac{\partial C}{\partial a_j^L} \frac{\partial a_j^L}{\partial z_j^L}$$

- Proof of BP2

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l}$$

- Proof of BP2 $\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l} = \sum_k \frac{\partial C}{\partial z_k^{l+1}} \frac{\partial z_k^{l+1}}{\partial z_j^l}$$

- Proof of BP2 $\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l} = \sum_k \frac{\partial C}{\partial z_k^{l+1}} \frac{\partial z_k^{l+1}}{\partial z_j^l} = \sum_k \frac{\partial z_k^{l+1}}{\partial z_j^l} \delta_k^{l+1}$$

- Proof of BP2 $\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l} = \sum_k \frac{\partial C}{\partial z_k^{l+1}} \frac{\partial z_k^{l+1}}{\partial z_j^l} = \sum_k \frac{\partial z_k^{l+1}}{\partial z_j^l} \delta_k^{l+1}$$

$$z_k^{l+1} = \sum_j w_{kj}^{l+1} a_j^l + b_k^{l+1}$$

- Proof of BP2 $\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l} = \sum_k \frac{\partial C}{\partial z_k^{l+1}} \frac{\partial z_k^{l+1}}{\partial z_j^l} = \sum_k \frac{\partial z_k^{l+1}}{\partial z_j^l} \delta_k^{l+1}$$

$$z_k^{l+1} = \sum_j w_{kj}^{l+1} a_j^l + b_k^{l+1} = \sum_j w_{kj}^{l+1} \sigma(z_j^l) + b_k^{l+1}$$

- Proof of BP2 $\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l} = \sum_k \frac{\partial C}{\partial z_k^{l+1}} \frac{\partial z_k^{l+1}}{\partial z_j^l} = \sum_k \frac{\partial z_k^{l+1}}{\partial z_j^l} \delta_k^{l+1}$$

$$z_k^{l+1} = \sum_j w_{kj}^{l+1} a_j^l + b_k^{l+1} = \sum_j w_{kj}^{l+1} \sigma(z_j^l) + b_k^{l+1}$$

$$\frac{\partial z_k^{l+1}}{\partial z_j^l} = w_{kj}^{l+1} \sigma'(z_j^l)$$

- Proof of BP2 $\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l} = \sum_k \frac{\partial C}{\partial z_k^{l+1}} \frac{\partial z_k^{l+1}}{\partial z_j^l} = \sum_k \frac{\partial z_k^{l+1}}{\partial z_j^l} \delta_k^{l+1}$$

$$z_k^{l+1} = \sum_j w_{kj}^{l+1} a_j^l + b_k^{l+1} = \sum_j w_{kj}^{l+1} \sigma(z_j^l) + b_k^{l+1}$$

$$\frac{\partial z_k^{l+1}}{\partial z_j^l} = w_{kj}^{l+1} \sigma'(z_j^l)$$

$$\delta_j^l = \sum_k w_{kj}^{l+1} \delta_k^{l+1} \sigma'(z_j^l)$$

- Proof of BP3

$$\frac{\partial C}{\partial b_j^l} = \delta_j^l$$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l}$$

- Proof of BP3

$$\frac{\partial C}{\partial b_j^l} = \delta_j^l$$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l}$$

$$\frac{\partial C}{\partial b_j^l} = \sum_k \frac{\partial C}{\partial z_k^l} \frac{\partial z_k^l}{\partial b_j^l}$$

- Proof of BP3

$$\frac{\partial C}{\partial b_j^l} = \delta_j^l$$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l}$$

$$\begin{aligned} \frac{\partial C}{\partial b_j^l} &= \sum_k \frac{\partial C}{\partial z_k^l} \frac{\partial z_k^l}{\partial b_j^l} \\ &= \frac{\partial C}{\partial z_j^l} \frac{\partial z_j^l}{\partial b_j^l} \end{aligned}$$

- Proof of BP3

$$\frac{\partial C}{\partial b_j^l} = \delta_j^l$$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l}$$

$$\begin{aligned} \frac{\partial C}{\partial b_j^l} &= \sum_k \frac{\partial C}{\partial z_k^l} \frac{\partial z_k^l}{\partial b_j^l} \\ &= \frac{\partial C}{\partial z_j^l} \frac{\partial z_j^l}{\partial b_j^l} \\ &= \frac{\partial C}{\partial z_j^l} = \delta_j^l \end{aligned}$$

- Proof of BP4

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l$$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l}$$

- Proof of BP4

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l$$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l}$$

$$\frac{\partial C}{\partial w_{jk}^l} = \sum_i \frac{\partial C}{\partial z_i^l} \frac{\partial z_i^l}{\partial w_{jk}^l}$$

- Proof of BP4

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l$$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l}$$

$$\frac{\partial C}{\partial w_{jk}^l} = \sum_i \frac{\partial C}{\partial z_i^l} \frac{\partial z_i^l}{\partial w_{jk}^l}$$

$$z_i^l = \sum_h w_{ih}^l \cdot a_h^{l-1} + b_i^l$$

- Proof of BP4

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l$$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l}$$

$$\frac{\partial C}{\partial w_{jk}^l} = \sum_i \frac{\partial C}{\partial z_i^l} \frac{\partial z_i^l}{\partial w_{jk}^l}$$

$$z_i^l = \sum_h w_{ih}^l \cdot a_h^{l-1} + b_i^l$$

if $i = j$ and $h = k$

$$\frac{\partial z_j^l}{\partial w_{jk}^l} = a_k^{l-1}$$

- Proof of BP4

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l$$

$$\delta_j^l = \frac{\partial C}{\partial z_j^l}$$

$$\frac{\partial C}{\partial w_{jk}^l} = \sum_i \frac{\partial C}{\partial z_i^l} \frac{\partial z_i^l}{\partial w_{jk}^l}$$

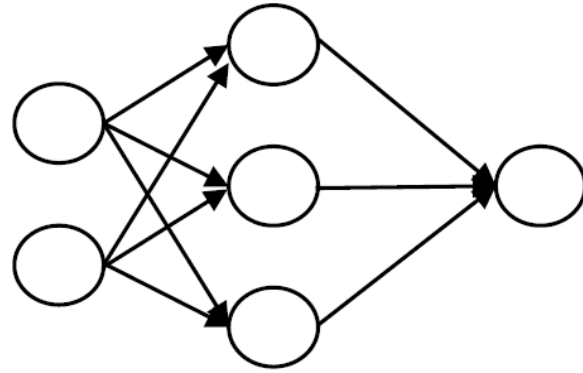
$$z_i^l = \sum_h w_{ih}^{l-1} \cdot a_h^{l-1} + b_i^l$$

if $i = j$ and $h = k$

$$\frac{\partial z_i^l}{\partial w_{jk}^{l-1}} = a_k^{l-1}$$

$$\frac{\partial C}{\partial w_{jk}^l} = \delta_j^l \cdot a_k^{l-1}$$

Below is a neural network:



All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{x}_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$\mathbf{y}_1 = \begin{bmatrix} 0 \end{bmatrix} \quad \mathbf{y}_2 = \begin{bmatrix} 0 \end{bmatrix} \quad \mathbf{y}_3 = \begin{bmatrix} 1 \end{bmatrix} \quad \mathbf{y}_4 = \begin{bmatrix} 0 \end{bmatrix}$$

And the learning rate is $\eta = 0.8$

We are processing 4 inputs at the same time by having all 4 of them as a matrix.

Round off all the number to the nearest 1000th. For example, 6.6543 is 6.654 and 3.3355 is 3.336.

z	0	0.5	1	1.5	2	2.5	3	3.5	4
a	0.5	0.622	0.731	0.818	0.881	0.924	0.953	0.971	0.982

z	1.111	1.122	1.211	1.231	1.308	1.318	1.411	1.422	1.511
a	0.752	0.754	0.770	0.774	0.787	0.789	0.804	0.806	0.819

And the back propagation equations are as follows:

Summary: the equations of backpropagation

$$\delta^L = \nabla_a C \odot \sigma'(z^L) \quad (\text{BP1})$$

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l) \quad (\text{BP2})$$

$$\frac{\partial C}{\partial b_j^l} = \delta_j^l \quad (\text{BP3})$$

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l \quad (\text{BP4})$$

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{x}_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$\mathbf{y}_1 = \begin{bmatrix} 0 \end{bmatrix} \quad \mathbf{y}_2 = \begin{bmatrix} 0 \end{bmatrix} \quad \mathbf{y}_3 = \begin{bmatrix} 1 \end{bmatrix} \quad \mathbf{y}_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(a) [4 Points] What is Z^2 matrix? First represent it with matrix format and write the result. Show all your work.

And the back propagation equations are as follows:

Summary: the equations of backpropagation

$$\delta^L = \nabla_a C \odot \sigma'(z^L) \quad (\text{BP1})$$

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l) \quad (\text{BP2})$$

$$\frac{\partial C}{\partial b_j^l} = \delta_j^l \quad (\text{BP3})$$

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l \quad (\text{BP4})$$

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{x}_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$\mathbf{y}_1 = \begin{bmatrix} 0 \end{bmatrix} \quad \mathbf{y}_2 = \begin{bmatrix} 0 \end{bmatrix} \quad \mathbf{y}_3 = \begin{bmatrix} 1 \end{bmatrix} \quad \mathbf{y}_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(a) [4 Points] What is Z^2 matrix? First represent it with matrix format and write the result. Show all your work.

$$\begin{aligned} Z^2 &= \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0.5 & 1.5 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 2 & 2 & 3 \end{bmatrix} \end{aligned}$$

(b) [4 Points] What is a^2 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(c) [4 Points] What is Z^3 matrix? First represent it with matrix format and write the result. Show all your work.

(b) [4 Points] What is a^2 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$$a^2 = \begin{bmatrix} 0.731 & 0.622 & 0.818 & 0.731 \\ 0.5 & 0.5 & 0.5 & 0.5 \\ 0.731 & 0.881 & 0.881 & 0.953 \end{bmatrix}$$

(c) [4 Points] What is Z^3 matrix? First represent it with matrix format and write the result. Show all your work.

(b) [4 Points] What is a^2 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$$a^2 = \begin{bmatrix} 0.731 & 0.622 & 0.818 & 0.731 \\ 0.5 & 0.5 & 0.5 & 0.5 \\ 0.731 & 0.881 & 0.881 & 0.953 \end{bmatrix}$$

(c) [4 Points] What is Z^3 matrix? First represent it with matrix format and write the result. Show all your work.

$$\begin{aligned} Z^3 &= \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} a^2 + \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1.231 & 1.122 & 1.318 & 1.231 \end{bmatrix} \end{aligned}$$

(d) [4 Points] What is a^3 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(e) [3 Points] What is ${}^1\delta^3$ value? ${}^1\delta^3$ is the layer 3 delta for input 1. Show all your work.

(d) [4 Points] What is a^3 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$$a^3 = \begin{bmatrix} 0.774 & 0.754 & 0.789 & 0.774 \end{bmatrix}$$

(e) [3 Points] What is δ^3 value? δ^3 is the layer 3 delta for input 1. Show all your work.

(d) [4 Points] What is a^3 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$$a^3 = [0.774 \quad 0.754 \quad 0.789 \quad 0.774]$$

(e) [3 Points] What is δ^3 value? δ^3 is the layer 3 delta for input 1. Show all your work.

$$\delta^3 = 0.774 \times (1 - 0.774) \times 0.774 = 0.135$$

(f) [3 Points] What is ${}^2\delta^3$ value? ${}^2\delta^3$ is the layer 3 delta for input 2. Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(g) [3 Points] What is ${}^3\delta^3$ value? ${}^3\delta^3$ is the layer 3 delta for input 3. Show all your work.

(f) [3 Points] What is ${}^2\delta^3$ value? ${}^2\delta^3$ is the layer 3 delta for input 2. Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$${}^2\delta^3 = 0.754 \times (1 - 0.754) \times 0.754 = 0.140$$

(g) [3 Points] What is ${}^3\delta^3$ value? ${}^3\delta^3$ is the layer 3 delta for input 3. Show all your work.

(f) [3 Points] What is ${}^2\delta^3$ value? ${}^2\delta^3$ is the layer 3 delta for input 2. Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$${}^2\delta^3 = 0.754 \times (1 - 0.754) \times 0.754 = 0.140$$

(g) [3 Points] What is ${}^3\delta^3$ value? ${}^3\delta^3$ is the layer 3 delta for input 3. Show all your work.

$${}^3\delta^3 = (0.789 - 1)(1 - 0.789) \times 0.789 = -0.035$$

(h) [3 Points] What is δ^3 value? δ^3 is the layer 3 delta for input 4. Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(i) [4 Points] What is average $\frac{\partial C}{\partial b^3}$ vector? Show all your work.

(h) [3 Points] What is δ^3 value? δ^3 is the layer 3 delta for input 4. Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$$\delta^3 = 0.774 \times (1 - 0.774) \times 0.774 = 0.135$$

(i) [4 Points] What is average $\frac{\partial C}{\partial b^3}$ vector? Show all your work.

(h) [3 Points] What is δ^3 value? δ^3 is the layer 3 delta for input 4. Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$$\delta^3 = 0.774 \times (1 - 0.774) \times 0.774 = 0.135$$

(i) [4 Points] What is average $\frac{\partial C}{\partial b^3}$ vector? Show all your work.

$$\begin{aligned} \frac{\partial C}{\partial b^3} &= (0.135 + 0.140 - 0.035 + 0.135) / 4 \\ &= 0.094 \end{aligned}$$

(j) [12 Points] What is average $\frac{\partial C}{\partial w^3}$ matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(j) [12 Points] What is average $\frac{\partial C}{\partial w^3}$ matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [0] \quad y_3 = [1] \quad y_4 = [0]$$

$$\begin{aligned} \frac{\partial C}{\partial w_{11}^3} &= (0.731 \times 0.135 + 0.622 \times 0.140 \\ &\quad - 0.818 \times 0.035 + 0.731 \times 0.135) / 4 \\ &= 0.064 \end{aligned}$$

$$\begin{aligned} \frac{\partial C}{\partial w_{12}^3} &= (0.5 \times 0.135 + 0.5 \times 0.140 - 0.5 \times 0.035 \\ &\quad + 0.5 \times 0.135) / 4 = 0.047 \end{aligned}$$

$$\begin{aligned} \frac{\partial C}{\partial w_{13}^3} &= (0.731 \times 0.135 + 0.881 \times 0.140 - 0.881 \times 0.035 \\ &\quad + 0.953 \times 0.135) / 4 = 0.080 \end{aligned}$$

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(k) [3 Points] What is new b^3 vector? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(k) [3 Points] What is new b^3 vector? Show all your work.

$$b^3 = \begin{bmatrix} 0 \end{bmatrix} - [0.8 \times 0.094] = \begin{bmatrix} -0.075 \end{bmatrix}$$

(I) [3 Points] What is new w^3 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(I) [3 Points] What is new w^3 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

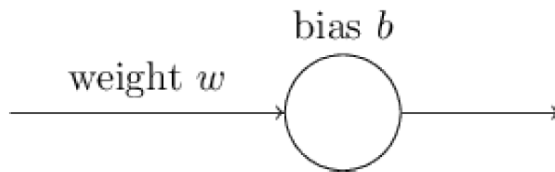
Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 0 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$$\begin{aligned} w^3 &= \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} - \\ &\quad \begin{bmatrix} 0.8 \times 0.064 & 0.8 \times 0.047 & 0.8 \times 0.086 \end{bmatrix} \\ &= \begin{bmatrix} 0.944 & 0.962 & -0.064 \end{bmatrix} \end{aligned}$$

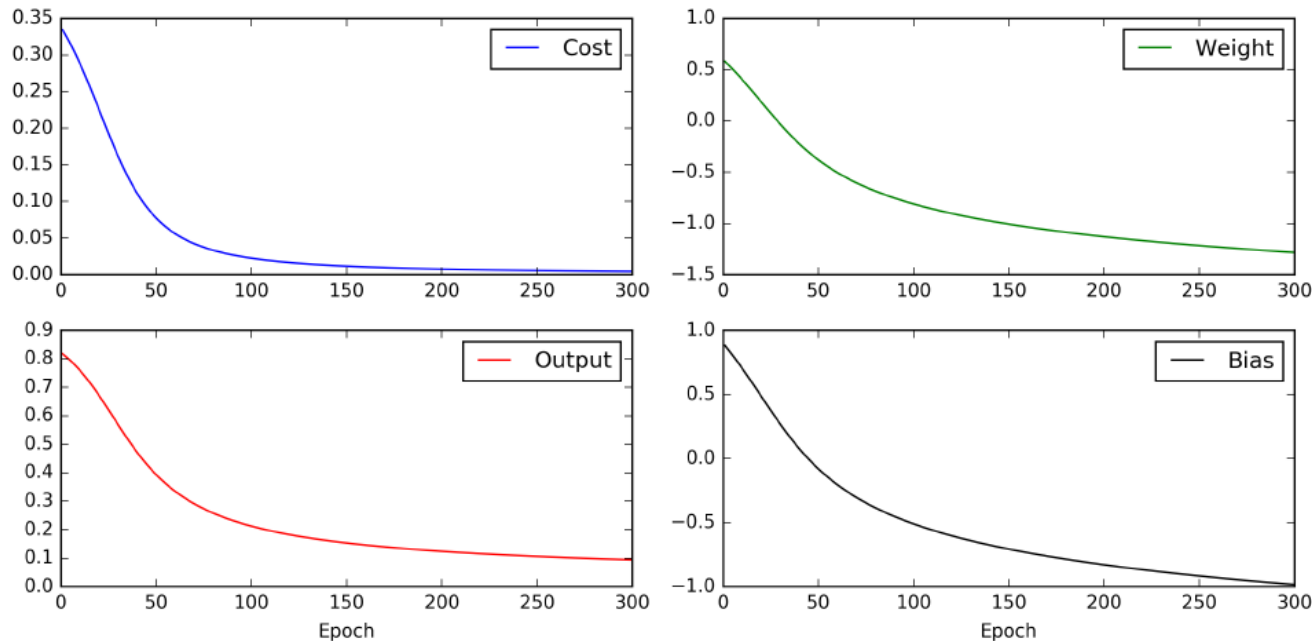
Cross-entropy cost function

- It'll be good if the neural network learns fast from it's errors
 - A toy network which take one input "1" and the desired output is "0"
 - Initial weight is "0.6" and initial bias is "0.9"
 - $z = "1.5"$ and $a = "0.82"$
 - $\eta=0.15$
 - Cost function is quadratic cost function



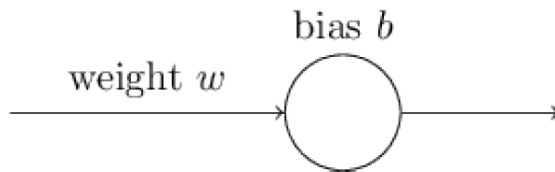
Cross-entropy cost function

- It'll be good if the neural network learns fast from it's errors
 - A toy network which take one input "1" and the desired output is "0"
 - Initial weight is "0.6" and initial bias is "0.9"
 - $z = "1.5"$ and $a = "0.82"$
 - $\eta = 0.15$
 - Cost function is quadratic cost function



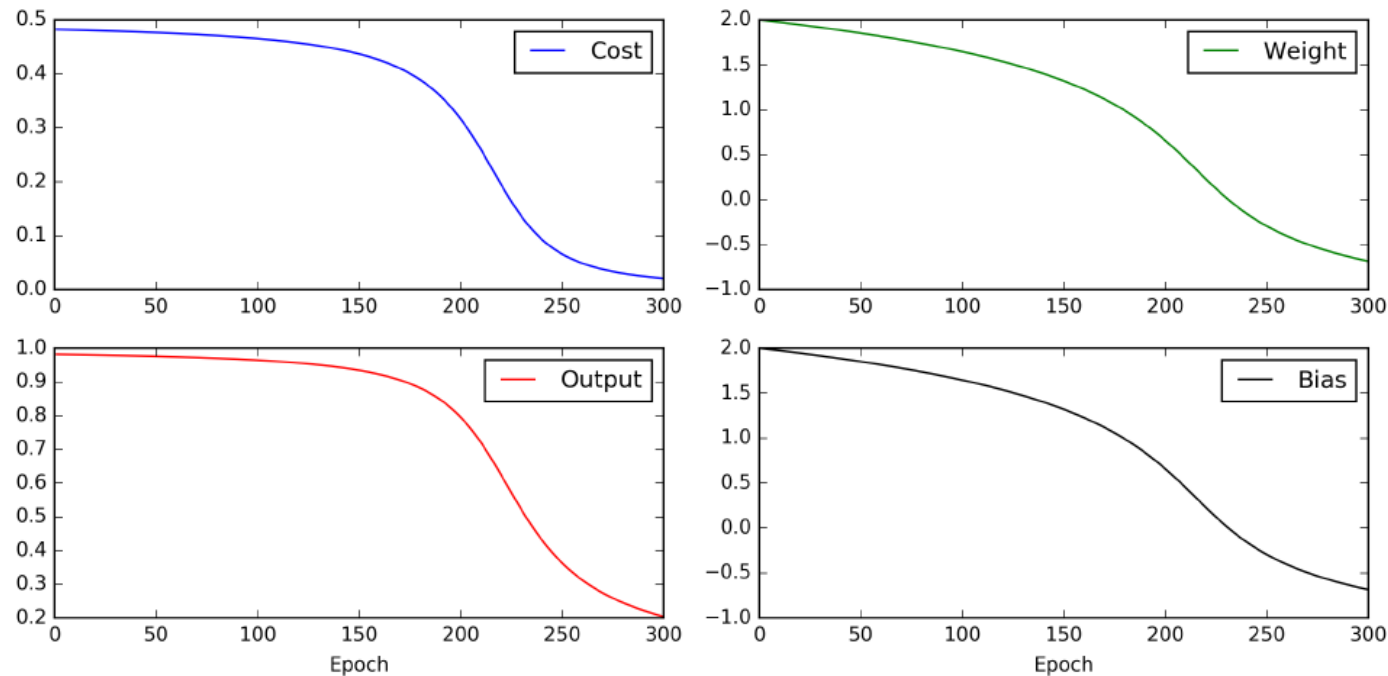
Cross-entropy cost function

- It'll be good if the neural network learns fast from it's errors
 - A toy network which take one input "1" and the desired output is "0"
 - Initial weight is "2.0" and initial bias is "2.0"
 - $z = "4"$ and $a = "0.98"$
 - $\eta=0.15$
 - Cost function is quadratic cost function



Cross-entropy cost function

- It'll be good if the neural network learns fast from it's errors
 - A toy network which take one input "1" and the desired output is "0"
 - Initial weight is "2.0" and initial bias is "2.0"
 - $z = "4"$ and $a = "0.98"$
 - $\eta = 0.15$
 - Cost function is quadratic cost function



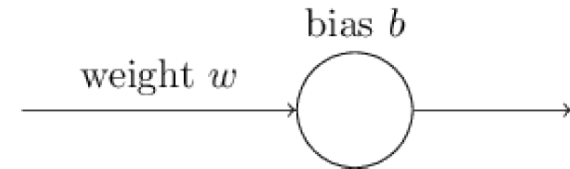
Cross-entropy cost function

- It'll be good if the neural network learns fast from it's errors
 - A toy network which take one input "1" and the desired output is "0"
 - Initial weight is "2.0" and initial bias is "2.0"
 - $z = "4"$ and $a = "0.98"$
 - $\eta=0.15$
 - Cost function is quadratic cost function

$$C = \frac{(y - a)^2}{2}$$

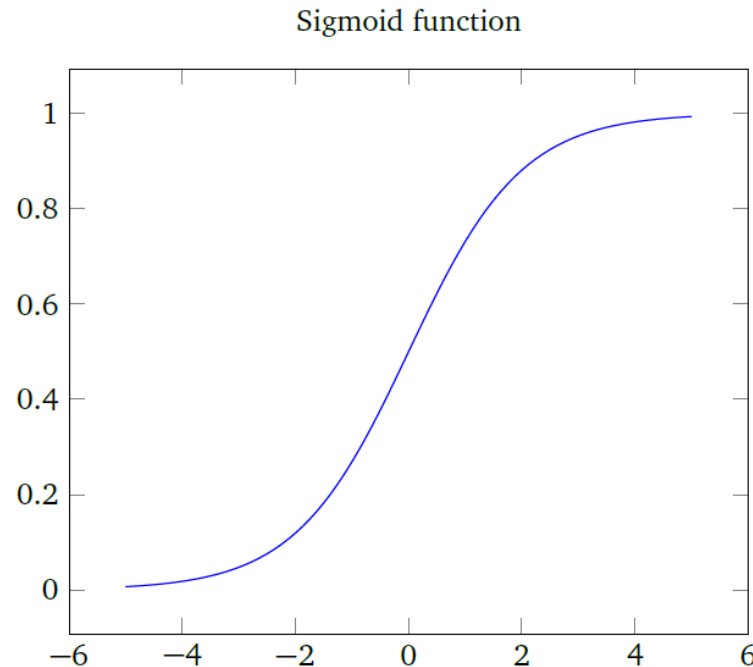
$$\frac{\partial C}{\partial w} = (a - y)\sigma'(z)x = a\sigma'(z)$$

$$\frac{\partial C}{\partial b} = (a - y)\sigma'(z) = a\sigma'(z),$$

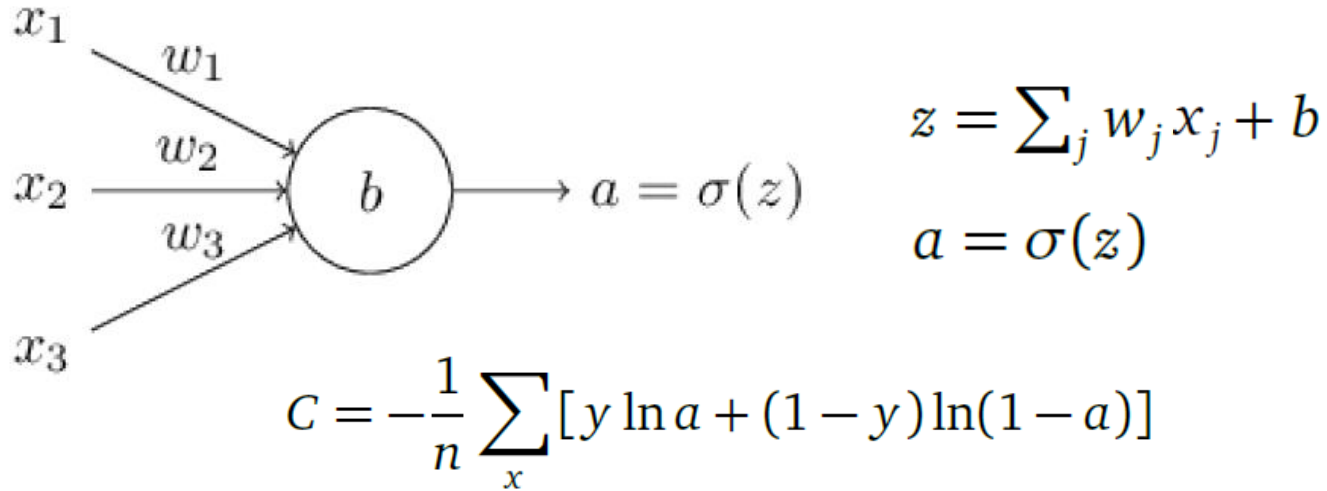


Cross-entropy cost function

- It'll be good if the neural network learns fast from it's errors
 - A toy network which take one input "1" and the desired output is "0"
 - Initial weight is "2.0" and initial bias is "2.0"
 - $z = "4"$ and $a = "0.98"$
 - $\eta = 0.15$
 - Cost function is quadratic cost function

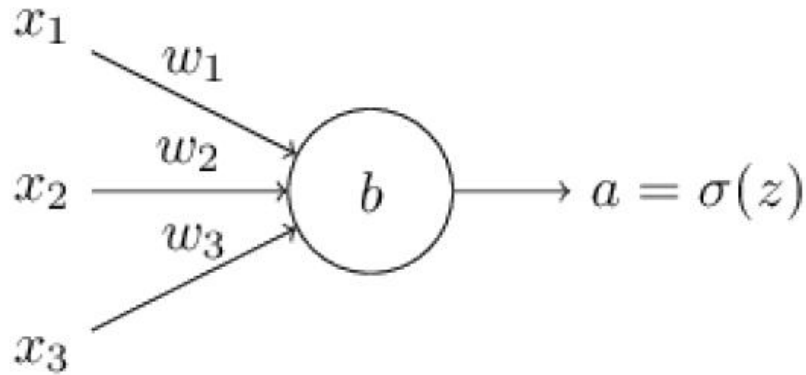


Cross-entropy cost function



Is this a Cost function?

Cross-entropy cost function



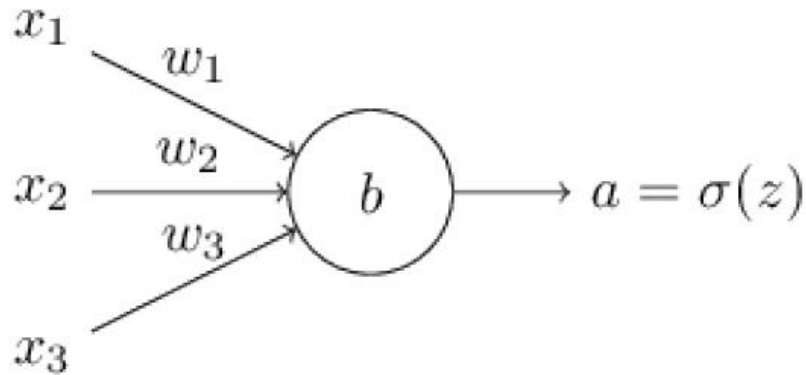
$$z = \sum_j w_j x_j + b$$

$$C = -\frac{1}{n} \sum_x [y \ln a + (1 - y) \ln(1 - a)]$$

Does it fix the learning slowdown problem?

$$\frac{\partial C}{\partial w_j} =$$

Cross-entropy cost function



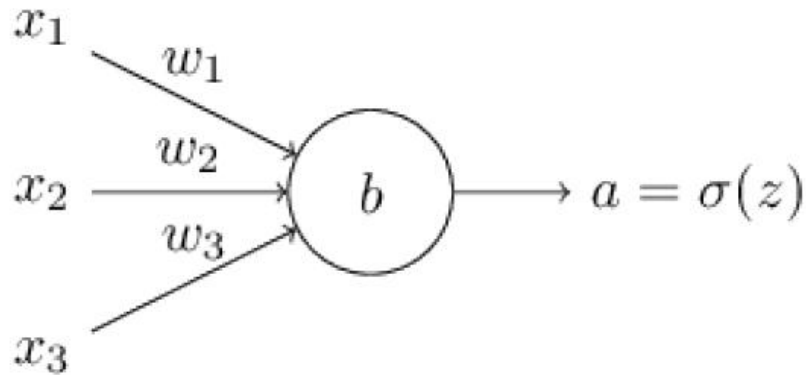
$$z = \sum_j w_j x_j + b$$

$$C = -\frac{1}{n} \sum_x [y \ln a + (1 - y) \ln(1 - a)]$$

Does it fix the learning slowdown problem?

$$\frac{\partial C}{\partial w_j} = -\frac{1}{n} \sum_x \left(\frac{y}{\sigma(z)} - \frac{1-y}{1-\sigma(z)} \right) \frac{\partial \sigma}{\partial w_j}$$

Cross-entropy cost function



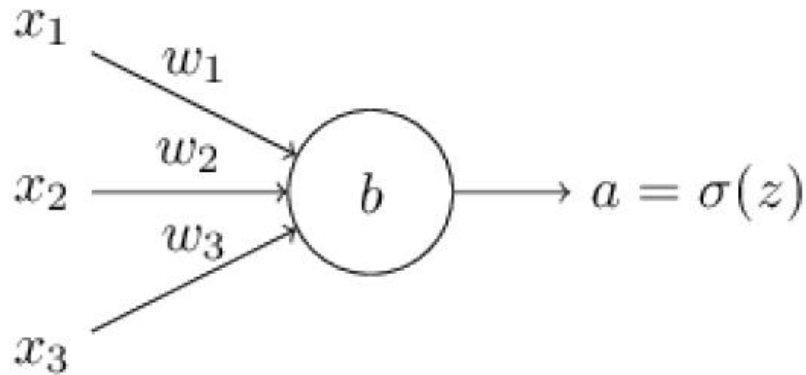
$$z = \sum_j w_j x_j + b$$

$$C = -\frac{1}{n} \sum_x [y \ln a + (1 - y) \ln(1 - a)]$$

Does it fix the learning slowdown problem?

$$\begin{aligned} \frac{\partial C}{\partial w_j} &= -\frac{1}{n} \sum_x \left(\frac{y}{\sigma(z)} - \frac{1-y}{1-\sigma(z)} \right) \frac{\partial \sigma}{\partial w_j} \\ &= -\frac{1}{n} \sum_x \left(\frac{y}{\sigma(z)} - \frac{1-y}{1-\sigma(z)} \right) \sigma'(z) x_j \end{aligned}$$

Cross-entropy cost function



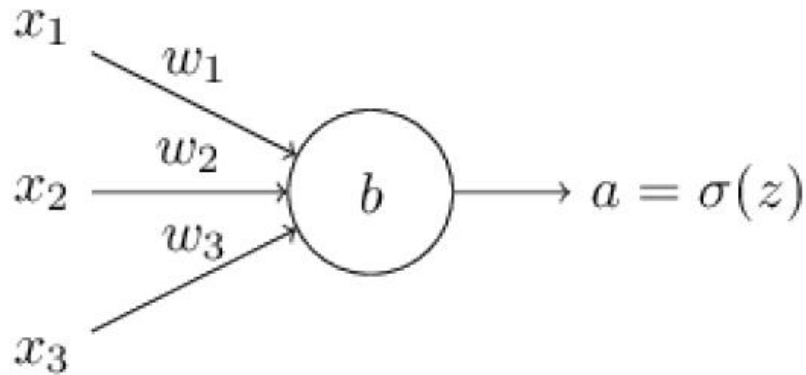
$$z = \sum_j w_j x_j + b$$

$$C = -\frac{1}{n} \sum_x [y \ln a + (1 - y) \ln(1 - a)]$$

Does it fix the learning slowdown problem?

$$\begin{aligned} \frac{\partial C}{\partial w_j} &= -\frac{1}{n} \sum_x \left(\frac{y}{\sigma(z)} - \frac{1-y}{1-\sigma(z)} \right) \frac{\partial \sigma}{\partial w_j} \\ &= -\frac{1}{n} \sum_x \left(\frac{y}{\sigma(z)} - \frac{1-y}{1-\sigma(z)} \right) \sigma'(z) x_j \\ &= \frac{1}{n} \sum_x \frac{\sigma'(z) x_j}{\sigma(z)(1-\sigma(z))} (\sigma(z) - y) \end{aligned}$$

Cross-entropy cost function



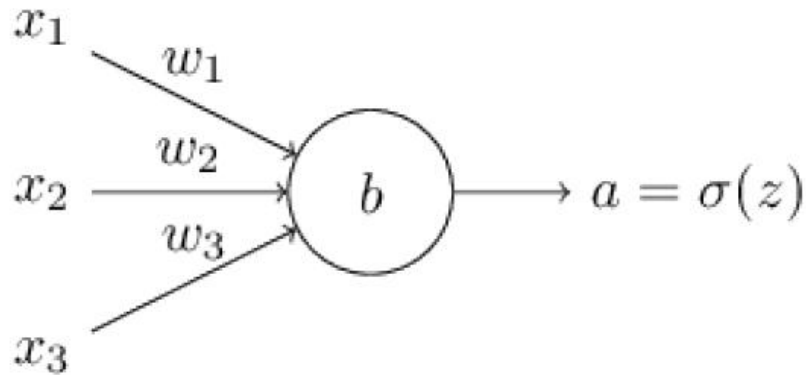
$$z = \sum_j w_j x_j + b$$

$$C = -\frac{1}{n} \sum_x [y \ln a + (1 - y) \ln(1 - a)]$$

Does it fix the learning slowdown problem?

$$\begin{aligned} \frac{\partial C}{\partial w_j} &= -\frac{1}{n} \sum_x \left(\frac{y}{\sigma(z)} - \frac{1-y}{1-\sigma(z)} \right) \frac{\partial \sigma}{\partial w_j} \\ &= -\frac{1}{n} \sum_x \left(\frac{y}{\sigma(z)} - \frac{1-y}{1-\sigma(z)} \right) \sigma'(z) x_j \\ &= \frac{1}{n} \sum_x \frac{\sigma'(z) x_j}{\sigma(z)(1-\sigma(z))} (\sigma(z) - y) \\ \frac{\partial C}{\partial w_j} &= \frac{1}{n} \sum_x x_j (\sigma(z) - y) \end{aligned}$$

Cross-entropy cost function



$$z = \sum_j w_j x_j + b$$

$$C = -\frac{1}{n} \sum_x [y \ln a + (1 - y) \ln(1 - a)]$$

Does it fix the learning slowdown problem?

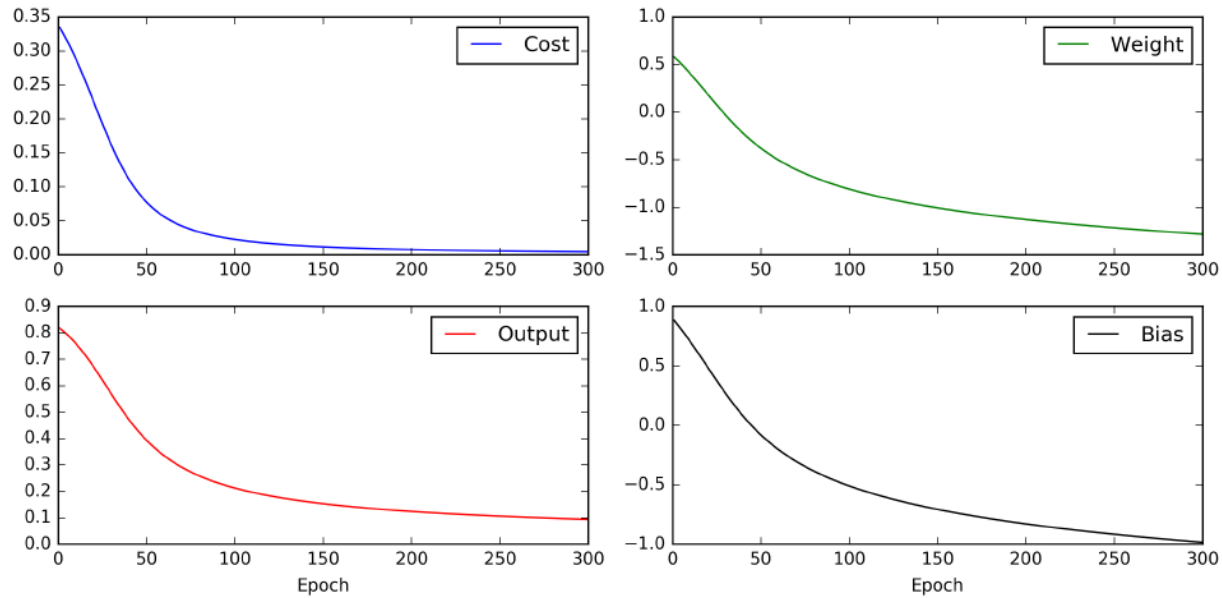
$$\begin{aligned} \frac{\partial C}{\partial w_j} &= -\frac{1}{n} \sum_x \left(\frac{y}{\sigma(z)} - \frac{1-y}{1-\sigma(z)} \right) \frac{\partial \sigma}{\partial w_j} \\ &= -\frac{1}{n} \sum_x \left(\frac{y}{\sigma(z)} - \frac{1-y}{1-\sigma(z)} \right) \sigma'(z) x_j \end{aligned}$$

$$= \frac{1}{n} \sum_x \frac{\sigma'(z) x_j}{\sigma(z)(1-\sigma(z))} (\sigma(z) - y)$$

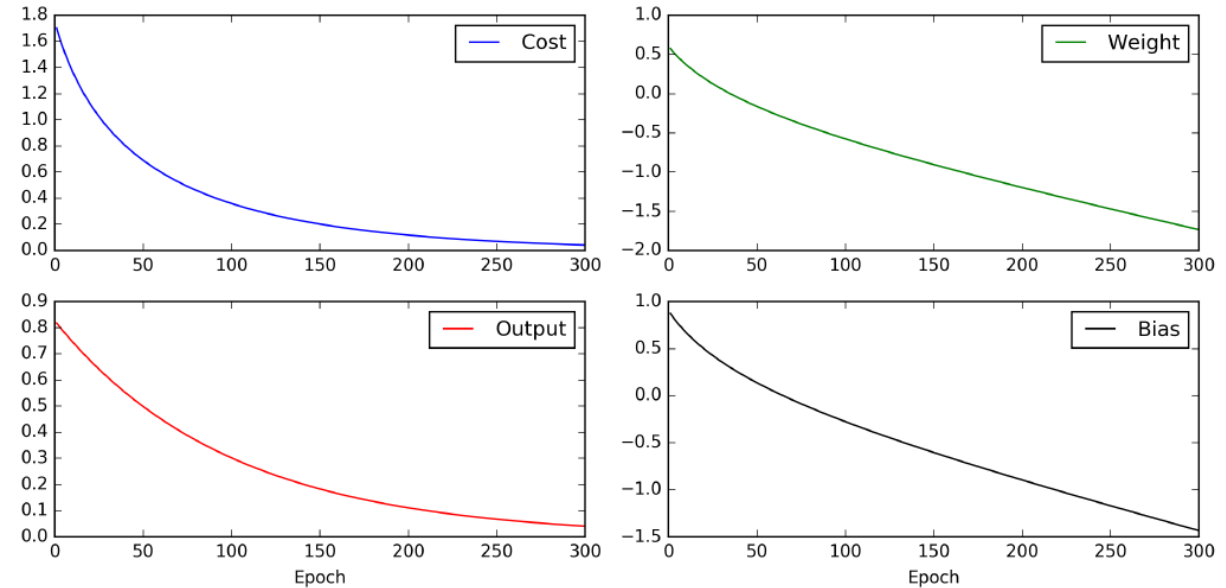
$$\frac{\partial C}{\partial w_j} = \frac{1}{n} \sum_x x_j (\sigma(z) - y) \quad \frac{\partial C}{\partial b} = \frac{1}{n} \sum_x (\sigma(z) - y)$$

Cross-entropy cost function

Quadratic cost function with $w=0.6$, $b=0.9$, $a=0.82$

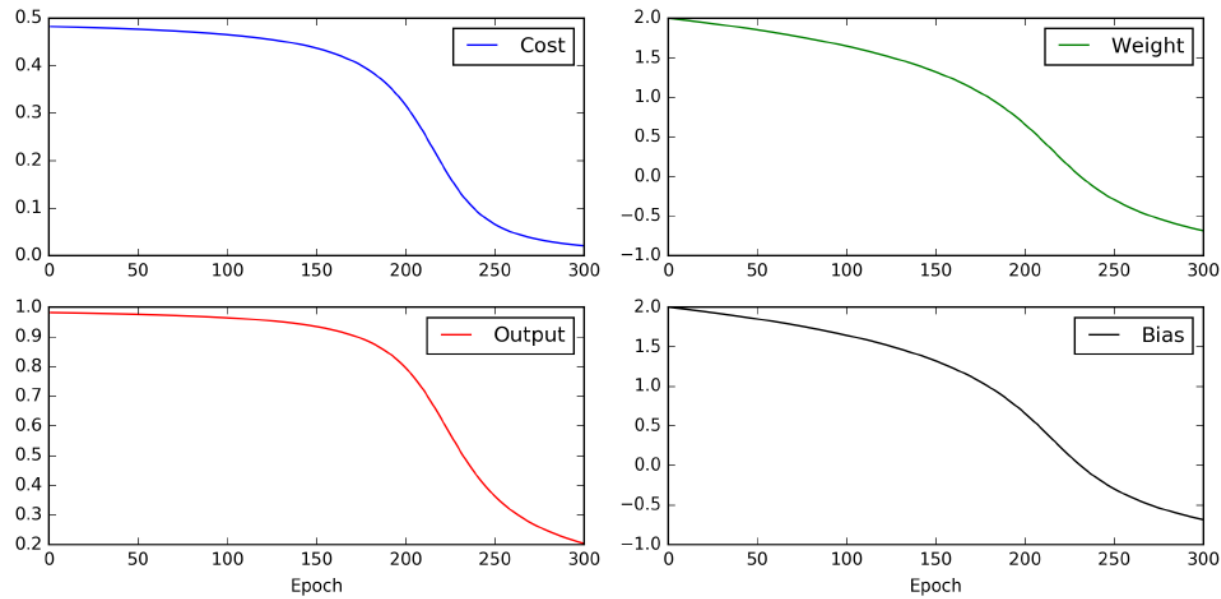


Cross-entropy cost function with $w=0.6$, $b=0.9$, $a=0.82$

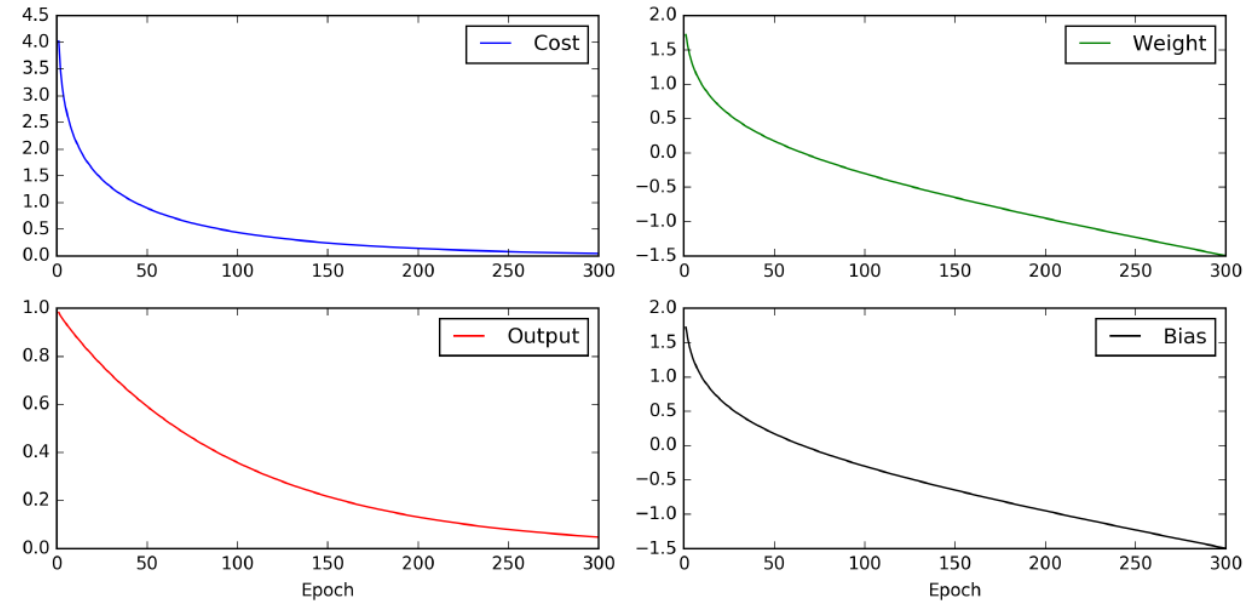


Cross-entropy cost function

Quadratic cost function with $w=b=2.0$, $a=0.98$



Cross-entropy cost function with $w=b=2.0$, $a=0.98$



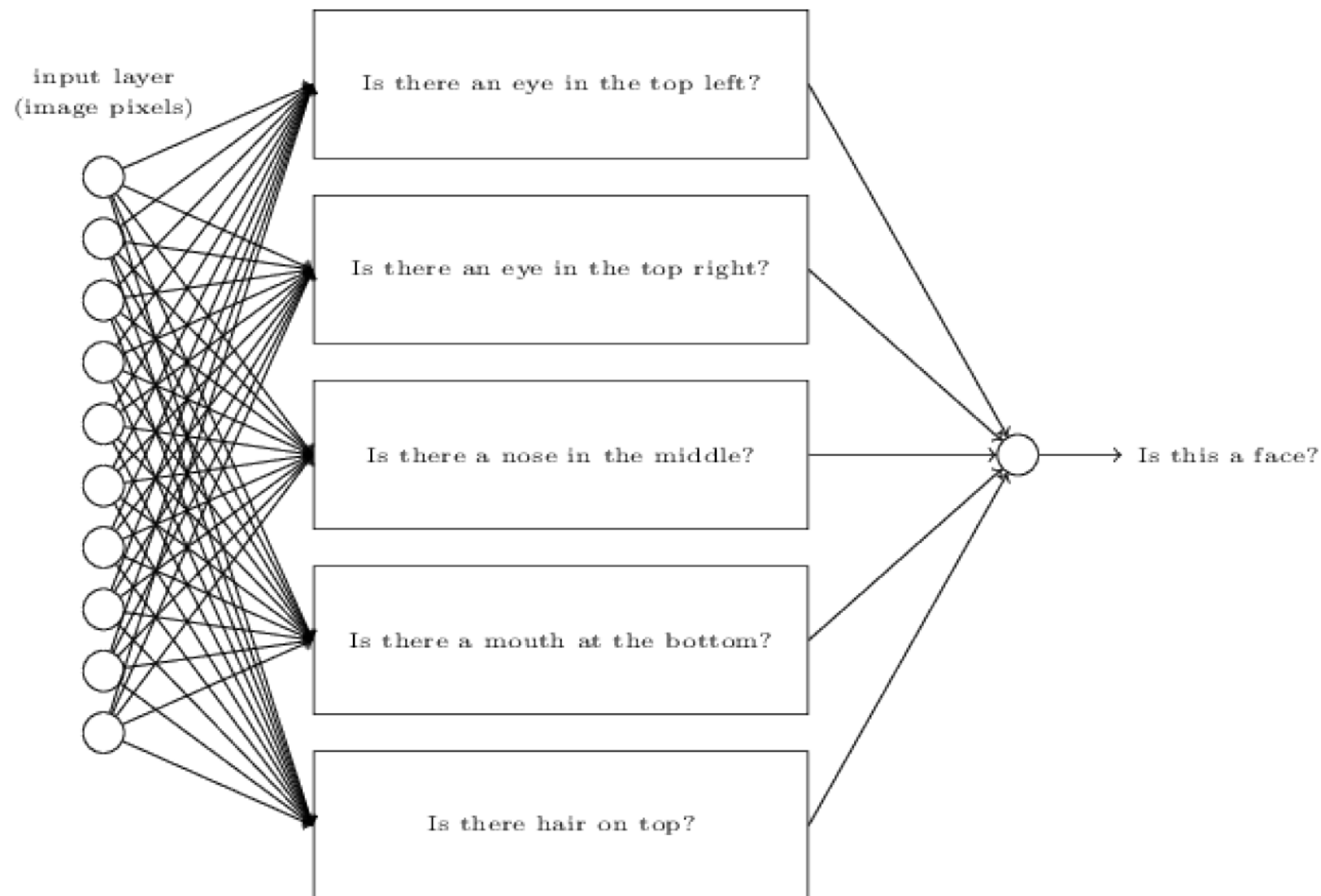
Toward Deep Learning

- **The weights and biases in the network are learned automatically**
 - **It's not easy to understand what each neuron does**
 - **Researchers have tried to understand the principles of how human brain and neural networks are working**
 - **We may not understand neither the brain nor the network works even in the future**

Toward Deep Learning

- **Think about a network recognizing human faces**
 - Input neurons take pixels as inputs
 - A single output neuron will indicate “Yes, it’s a face” or “No, it’s not a face”
- **Decompose a problem into subproblems**
 - Does it have an eye in the top right?
 - Does it have a nose in the middle?
 - Does it have a mouth in the bottom middle?
 - Is there hair on top?
 - If the answers to above questions are “yes” or just “probably yes” then the output neuron will indicate “Yes, it’s a face”

Toward Deep Learning



Toward Deep Learning

- **“Is there an eye in the top left?” can be decomposed into questions such as**
 - **“Is there an eyebrow?”**
 - **“Are there eyelashes?”**
 - **“Is there an iris?”**

Toward Deep Learning

